

An evaluation of the impact of end-to-end query optimization strategies on energy consumption

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E4C



Outline

- 1 Proposal: Evaluating the impact of end-to-end query optimization strategies on energy consumption
- 2 Results of Study
- 3 Threats to validity, Conclusion and Future work

Context

Evaluation of the energy consumed by Relational Database Systems

- Software applications intensively receive and transfer data by using a Relational Database Management Systems (*RDBMS*).
- RDBMS are software components that could require a lot of hardware resources querying large databases.
- Resources are limited and expensive so they must be used wisely. However, developers are not aware of the resource impact they cause when building software.



Figure: Energy is expensive



Figure: Applications require more energy every time

Problem and Research objectives

Problem

Studies are traditionally interested in reducing execution time by implementing databases optimizations strategies such as index or compression. **However, there are no studies dedicated to analyse the end-to-end (server and client sides) energy impact of these strategies.**

Research objectives

Our goal is to increase awareness about the **end-to-end energy impact** of databases optimizations strategies. It can serve as guidelines for supporting developers to choose the strategies that are more energy friendly.

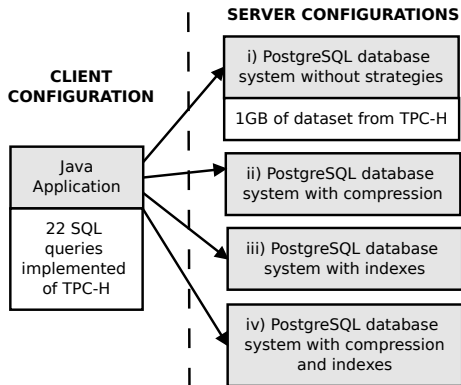
- RQ1: Does the implementation of database optimization strategies affect the energy consumption of client server applications?
- RQ2: Does the addition of database optimization strategies affect the energy consumption of client server applications?

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Evaluating the impact of end-to-end query optimization strategies on energy consumption

Design of the Study: Setup



Evaluating the impact of end-to-end query optimization strategies on energy consumption

Design of the Study: Algorithm

Require: *lstOpt*; // base, index, compression, index-compression

numIterations = 30; // number of iterations

lstQueries; // list of queries (22 queries en total)

Ensure: collect metrics for the analysis of energy consumption of a client-server application

```
1: for optimization in lstOpt do
2:   delete the database if it already exists
3:   create the database according to the configuration of optimization
4:   for query in lstQueries do
5:     for i in numIterations do
6:       restart the service of database server
7:       start monitoring tools
8:       execute query in the database server with the config.
9:       collect metrics (execution time from client, energy consumption from client and server)
10:      stop monitoring tools
11:    end for
12:  end for
13: end for
```

Evaluating the impact of end-to-end query optimization strategies on energy consumption

Juliet: Minimalist and Energy friendly tool, developed in C++

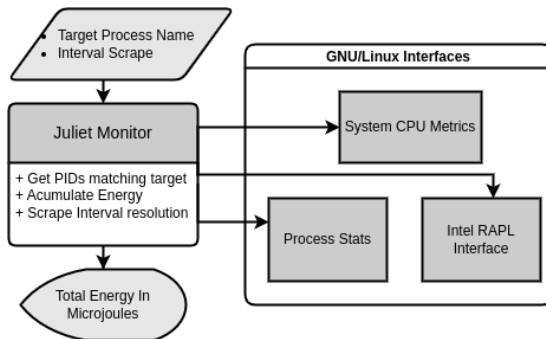


Figure: Juliet Architecture Design

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Results of Study

RQ1: Does the implementation of database optimization strategies affect the energy consumption of client server applications?

RQ2: Does the addition of database optimization strategies affect the energy consumption of client server applications?

Table: Sample Query Results. **AEC:** Average of energy consumption. **RSEC:** Relative standard deviation. **DEC:** Difference of EC vs Base

Query 22						
	Client			Server		
	AEC	RSEC	%DEC	AEC	RSEC	%DEC
Base	119.39	0.06	-	4427.2	0.07	-
Index	16.22	0.07	86.41	1.41	0.19	99.97
Compression	118.93	0.06	-	4387.18	0.07	-
Index-Compression	16.26	0.05	86.38	1.2	0.19	99.97

Results of Study

RQ1: Does the implementation of database optimization strategies affect the energy consumption of client server applications?

RQ2: Does the addition of database optimization strategies affect the energy consumption of client server applications?

Table: Sample Query Results. **AEC:** Average of energy consumption. **RSEC:** Relative standard deviation. **DEC:** Difference of EC vs Base

Query 17						
	Client			Server		
	AEC	RSEC	%DEC	AEC	RSEC	%DEC
Base	43.78	0.08	-	393.82	0.03	-
Index	22.74	0.09	48.06	241.73	0.01	38.62
Compression	23.76	0.08	45.73	265.73	0.02	32.53
Index-Compression	23.39	0.10	46.56	238.51	0.02	39.44

Results of Study

RQ1 and RQ2

RQ1

- The **index** strategy reduce between [1.15, 103.17] J for the client side energy consumption, whilst for the **compression** was [1.28, 20.02] J.
- For the server side, the **index** strategy reduce between [1.66, 4425.79] J, whilst the **compression** reduced between [1.21, 128.08] J.

Conclusion: $I > C$

RQ2

- For the **index-compression** strategy the reduction of the energy consumption was between [0.92, 86.38] J for the client side, whilst for the server side was between [1.33, 4426] J. These values are similar to the values obtained for the **index** strategy.

Conclusion: $I + C \equiv I$

Results of Study

Some Exceptions:

Query 05						
	Client			Server		
	AEC	RSEC	%DEC	AEC	RSEC	%DEC
Base	18.78	0.13	-	21.43	0.06	-
Index	18.32	0.11	-	26.50	0.07	-23.68
Compression	18.12	0.12	-	18.18	0.1	15.13
Index-Compression	19.14	0.11	-	25.04	0.08	-16.84

```
SELECT n_name, sum(l_extendedprice * (1 - l_discount)) as revenue
FROM customer, orders, lineitem, supplier, nation, region
WHERE c_custkey = o_custkey
AND l_orderkey = o_orderkey
AND l_suppkey = s_suppkey
AND c_nationkey = s_nationkey
AND s_nationkey = n_nationkey
AND n_regionkey = r_regionkey
AND r_name = 'ASIA'
AND o_orderdate ≥ date '1993 - 01 - 01'
AND o_orderdate < date '1993 - 01 - 01' + interval '1' year
GROUP BY n_name
ORDER BY revenue DESC
LIMIT 1;
```

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Threats to validity, Conclusion and Future work (1/2)

Threats to validity

- **Internal validity:** Juliet could introduce bias.
- **Construct validity:** Same computer used for running the client and server.
- **External validity:** Benchmarking limited to 22 specific queries, our results cannot be generalized.

Conclusion

- **Indexation tends to be more effective than data compression to reduce the energy** consumed by the execution of the majority of the queries (16/22 queries). The addition of indexation and compression does not lead to a better result comparing with indexation.
- **More experiments are needed to understand the reason of why some queries do not follow this tendency:** i) 3/22 queries increase energy consumption, and ii) 3/22 queries do not show any effect.

Future work

- Run new experiments in a real distributed software architecture.
- Run new experiments over other RDBMS and NoSQL datasets.
- Classify the queries according their complexity, responses time and size to run new experiments.
- Consider the trade-off of energy consumption of database optimization strategies with other resources such as CPU, I/O usage, and memory.
- Run new experiments to compare Juliet with other tools like Scaphandre or PowerAPI.

Thank you!

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